

SHOBOT
a Tennis Ball Collecting Robot for Tournament Games
by Team ECHO

Oscar Chan
Charles Hung
Eito Ogawa
Hengze Li

Group
#18

PROJECT OVERVIEW

The team was tasked to develop a design for a **battery-operated vacuuming robot** that can collect tennis balls from the court, as a replacement for ball kids.

CLIENT SPECIFICATION

- ❑ To **vacuum and store** tennis balls
- ❑ **Aesthetically pleasing** design
- ❑ Production run of **10000 products**
- ❑ On and off court **quickly (~10s)**
- ❑ **Sustainable** design
→ <500 kg CO₂-eq per five years

KEY FEATURES

- ❖ Suction gripper to capture ball
- ❖ Stores 2-3 tennis balls
- ❖ Product life of 10 years

WHY SUCTION GRIPPERS?

Imagine using a “vacuum cleaner” robot running during a tennis match, it would...

Require a lot of power

Large motor size

Loud vacuum noise

- Very distracting and battery draining
- Suction cups fixes all three issues!

LINEAR ACTUATION MODULE

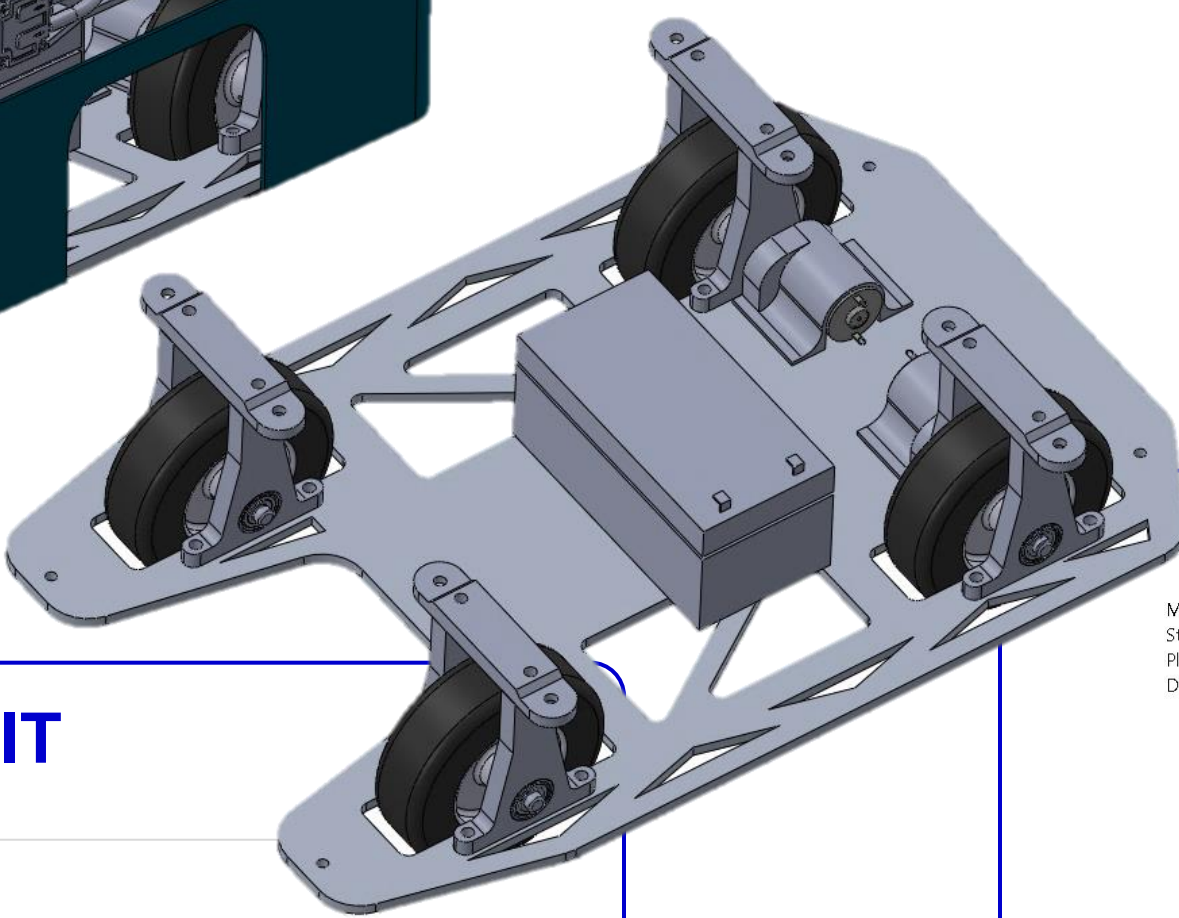
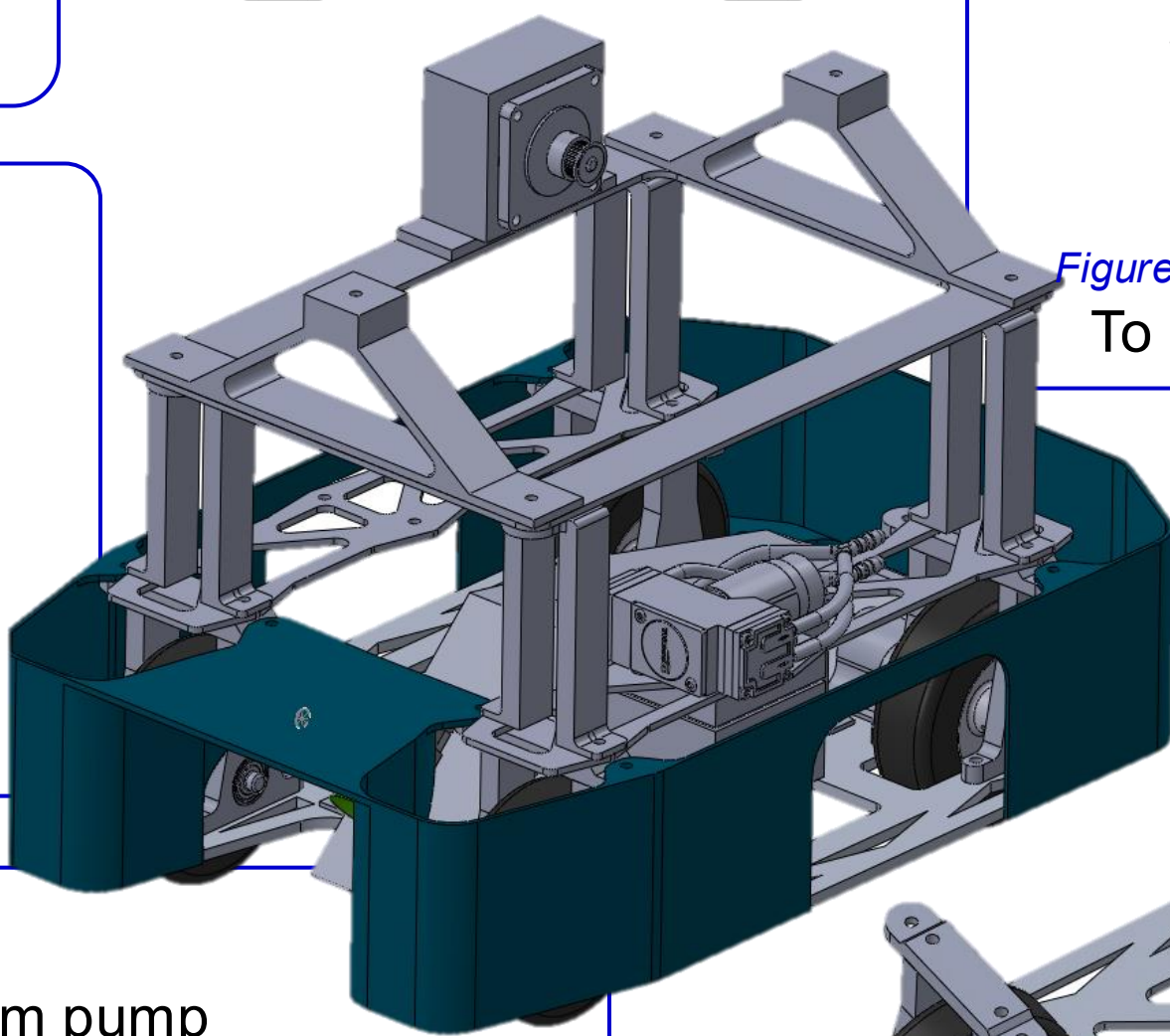
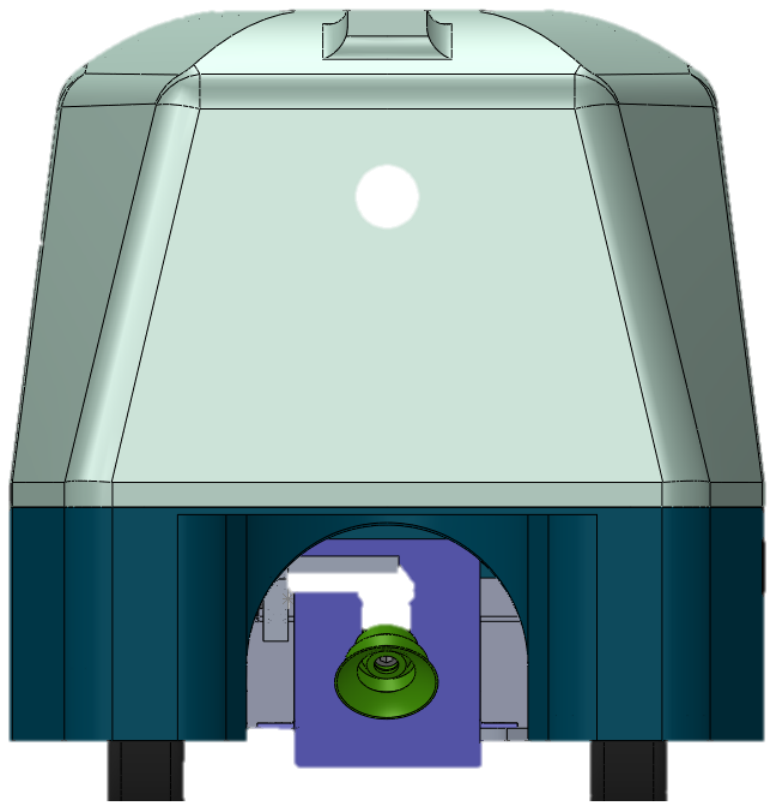
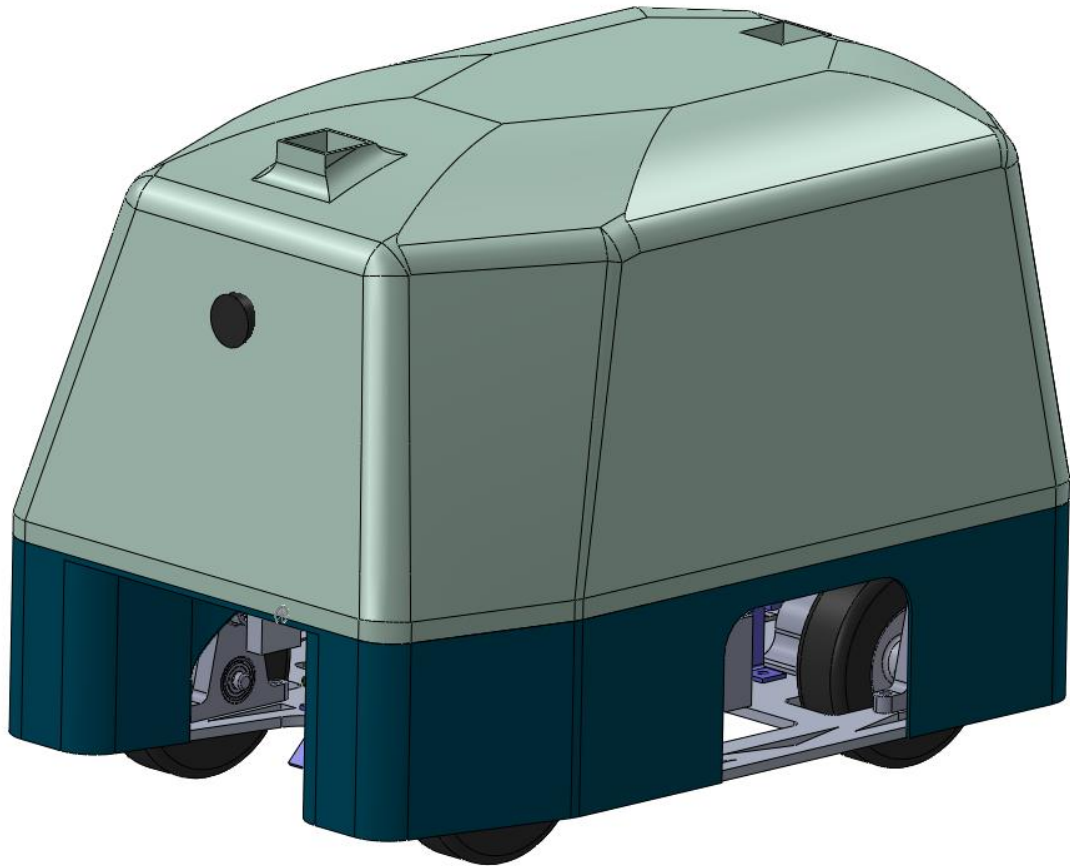
The linear actuator drags the suction cup diagonally up a ramp. It consists of:

- Stepper motor
- Timing belt/pulley
- Timing belt clamp
- Suction Cup Connector

VACUUM PUMP

- Instead of a centrifugal pimp, diaphragm pump
- can maintain a stable -ve pressure even with small leakage.
- No overheating under restricted inlet

OUR DESIGN SHOBOT



DRIVE TRANSMISSION

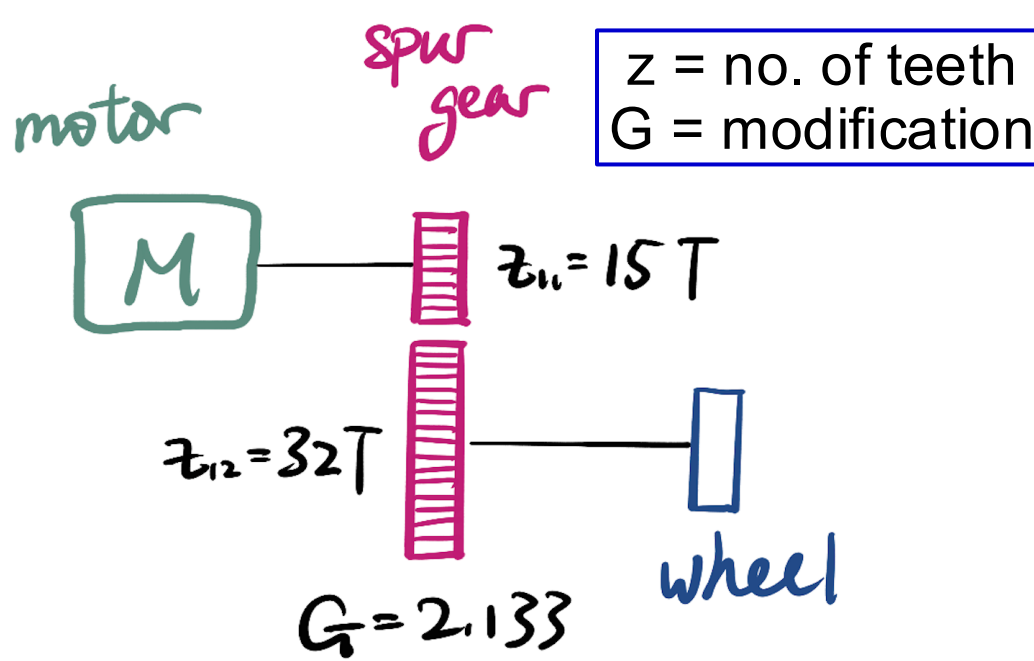


Figure 2: Drive transmission schematic diagram
To achieve a speed of 9.97 km/h

USER OPERATION

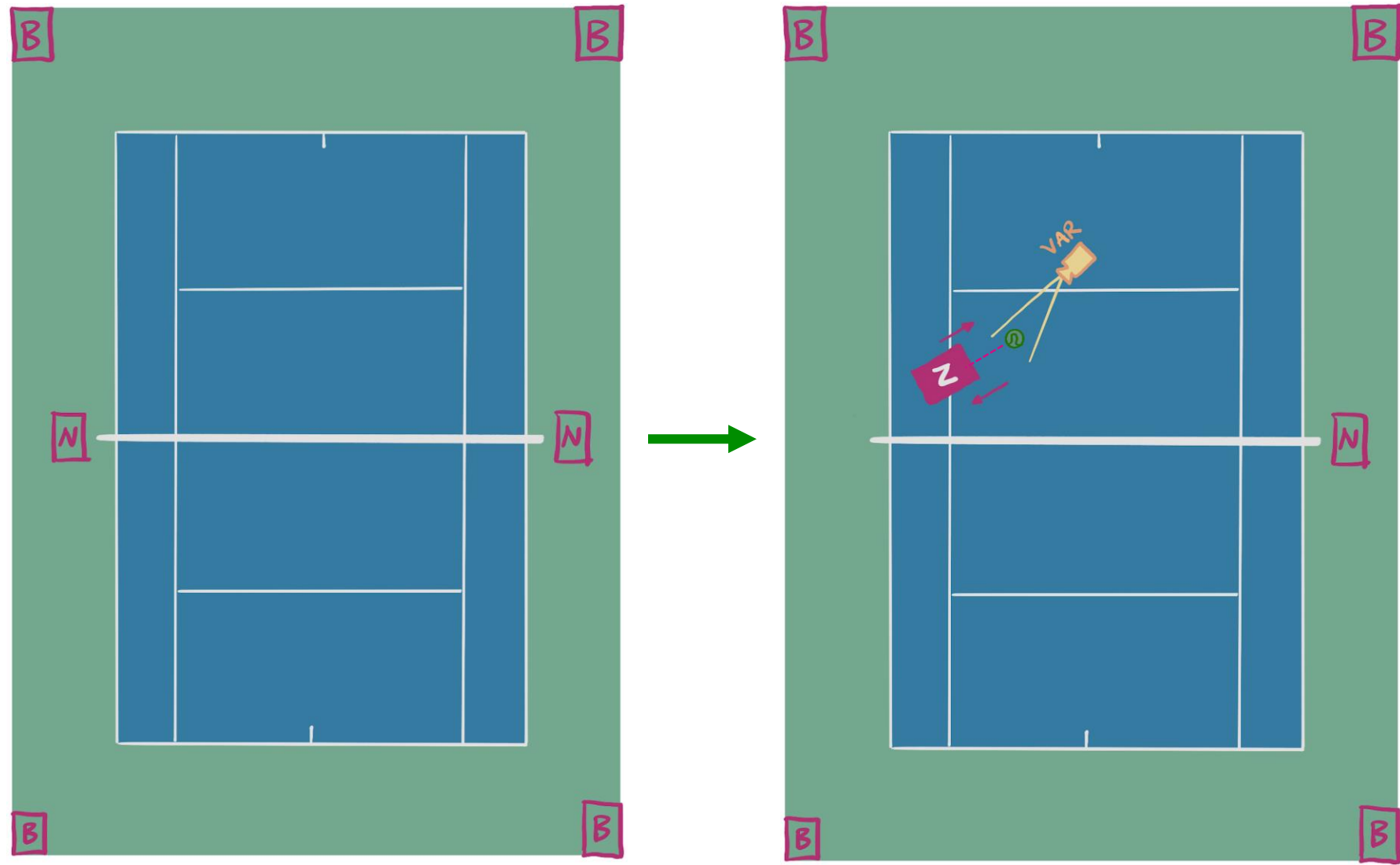


Figure 1: Tennis ball pickup process in macro view.

- Fully autonomous → Sensors and control by Electronics team
- One robot in each corner (Base), one by each side net (Net)
- Global ball tracking by VAR; local by LiDAR/camera sensors

MATERIAL SELECTION

Selected using Ansys Granta Edupack software.

Key Components	Materials	Manufacturing Method
Shaft	AISI 4140 [low-alloy steel]	CNC turning
Outer casing	ASA [thermoplastic]	Injection molding
Structural frame [chassis, struts]	A356-T6 [aluminium]	CNC milling

Table 1: Materials component selection

BATTERY SELECTION

12V, 3.2Ah Lead Rechargeable Battery

- For every 1000 ball captures (typical tennis game), uses ~0.8 Ah.
- Battery life = 3-4 tennis games

FAILURE MITIGATION

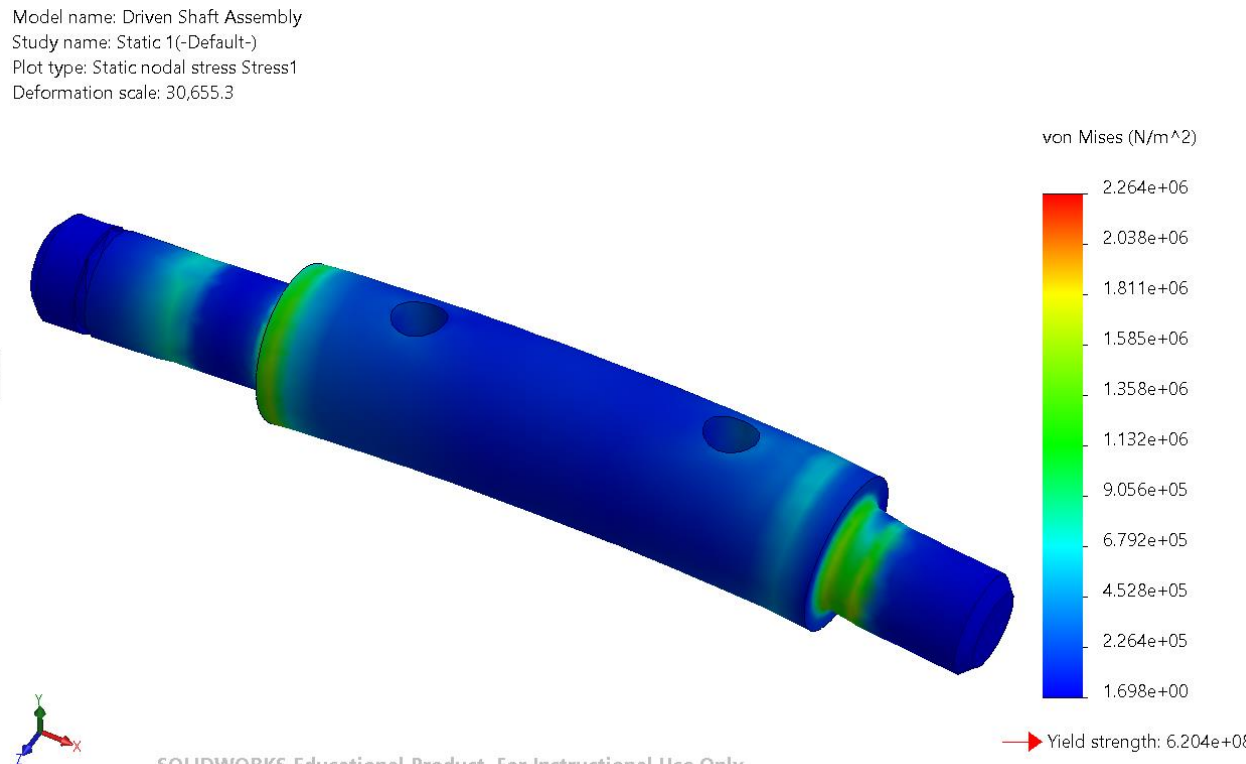


Figure 3: FEA of Wheel Shaft

SUSTAINABILITY INSIGHT: ECO-AUDIT

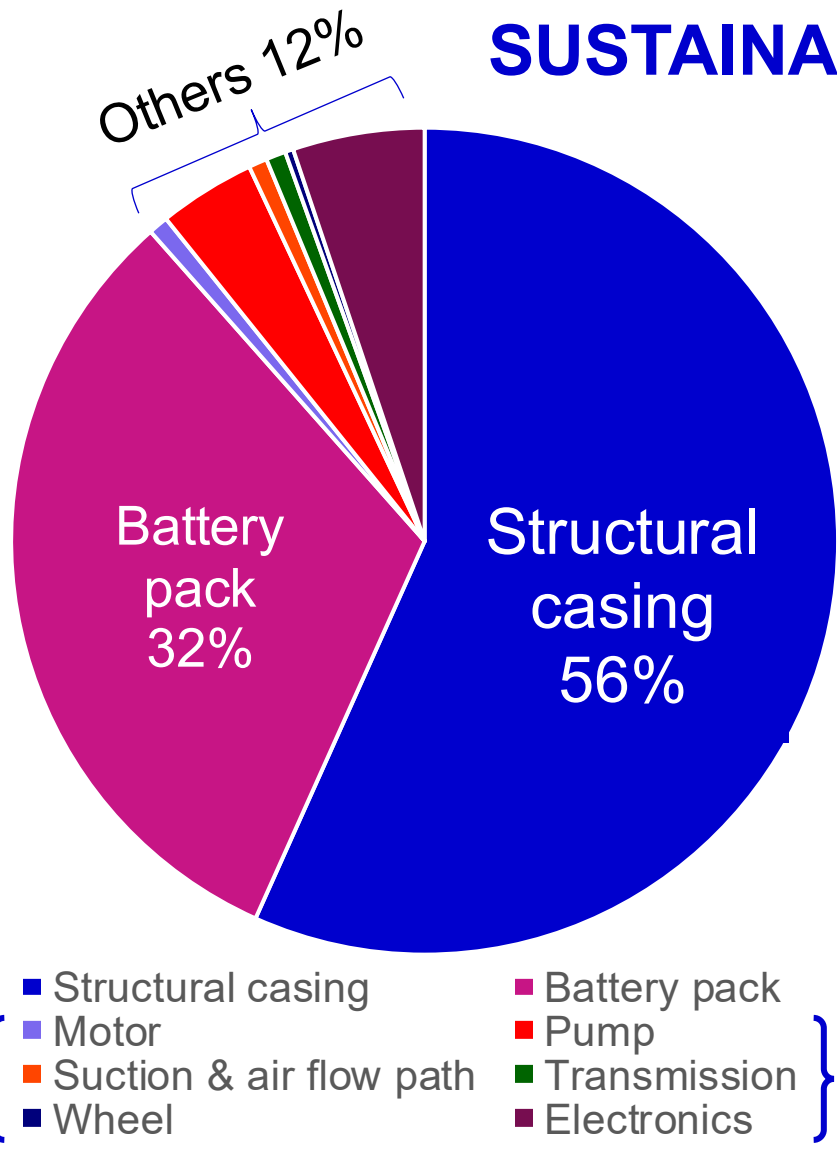


Figure 4: Carbon footprint (kg) by component mass.

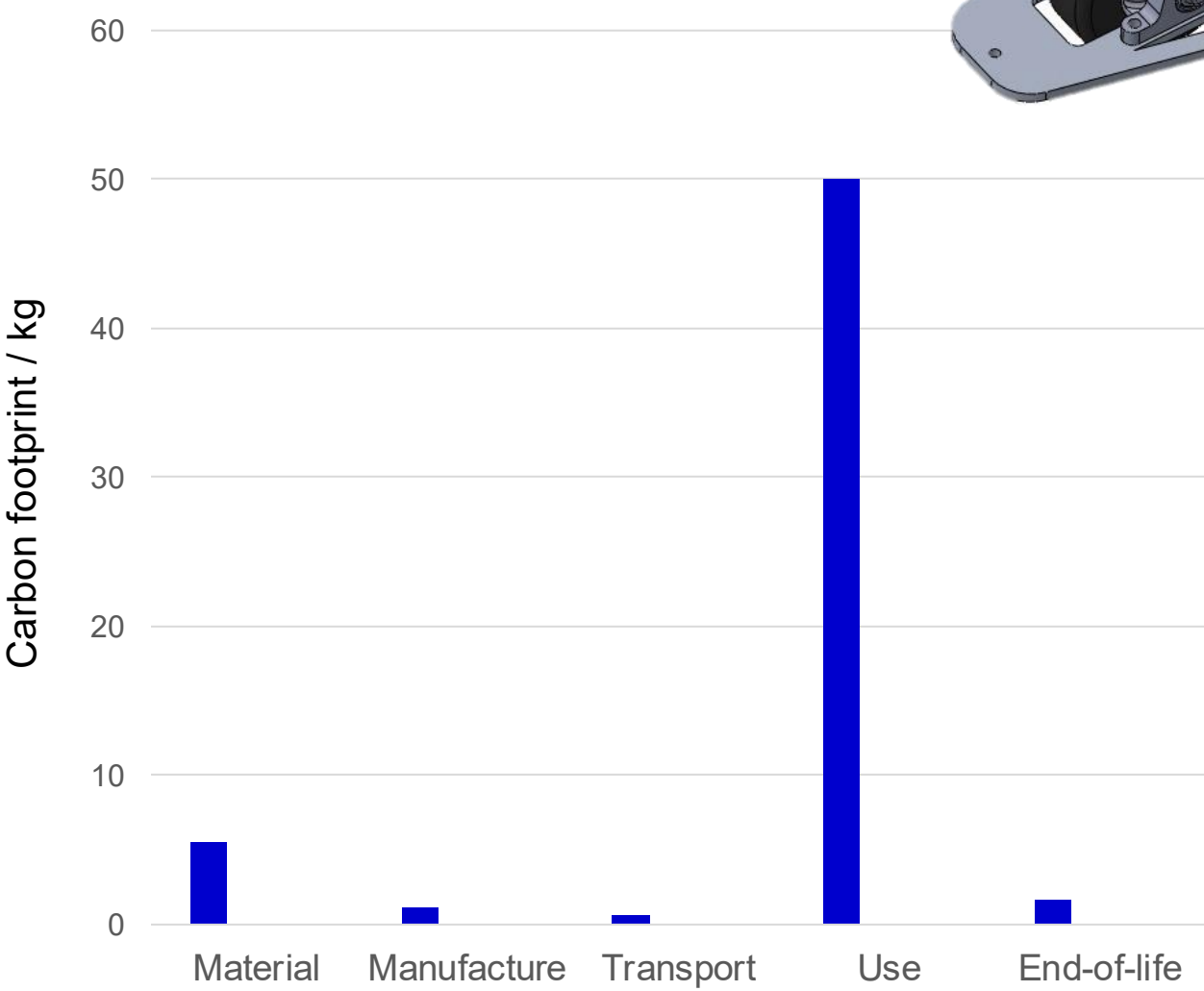


Figure 5: Carbon Footprint (kg) in product life cycle (10 years).

- Total CO₂-eq 58 kg per 10 years → achieves sustainability target

CONCLUSION

A battery-operated vacuuming robot is successfully designed, meeting all client specification requirements.

- **Modular** – layered design for ergonomics
- **Sustainable** – CO₂ footprint minimised
- **Non-disruptive** – 10s pick up time, no loud noise, aesthetic
- **Reliable** – Safety factor of shaft is 28.04